Synthetic Native Template

Redistributable test fixture — not laboratory data

Representative content layout

- Native title/content placeholders

Progress / Previous Commitments

Commitment | NS101; NS201

Current | H002

Parallel | interface-mechanism; interface-mechanism

H001 ｜ Hypothesis

Hypothesis ｜ Bulk conductivity drives the positional signal gradient.
Falsifier ｜ Matched conditions do not change the predicted CV or resistance.
Question ｜提升 bulk conductivity 是否足以降低位置依賴的訊號變異？

H001 ｜ Problem

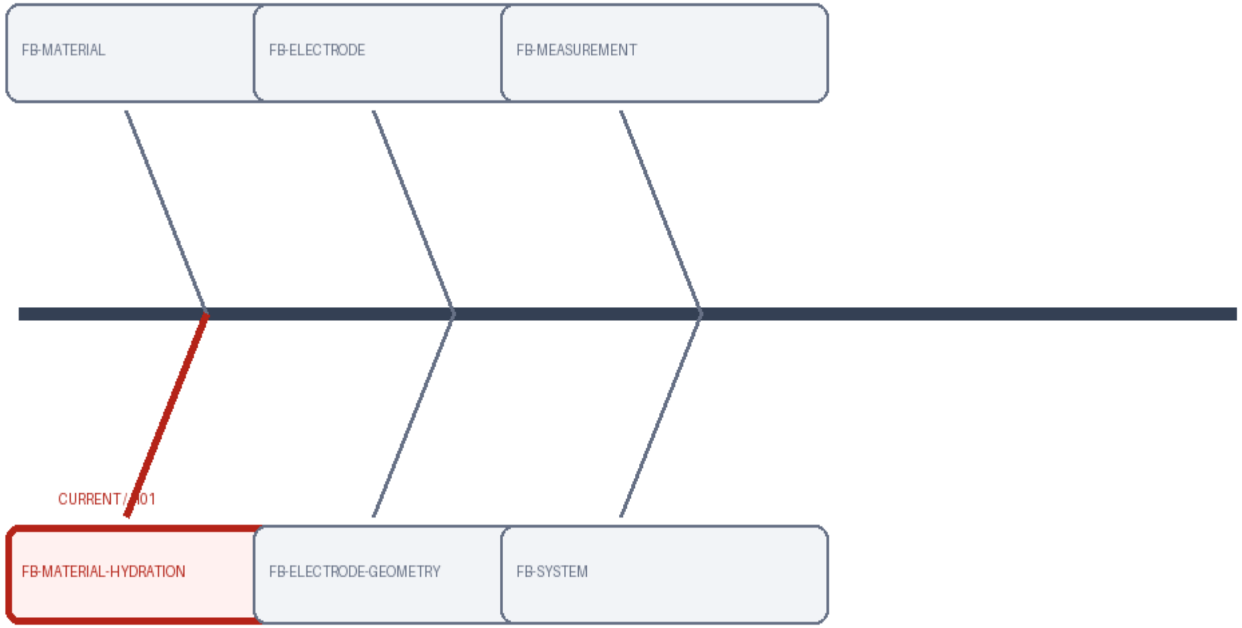
Prior ｜含水量與導電度存在位置梯度

Conflict ｜導電度提高是否足以修復重複性仍未知。

Question ｜提升 bulk conductivity 是否足以降低位置依賴的訊號變異？

H001 | Total Fishbone / Research Map

History | FB001 rev1
Focus | FB-MATERIAL-HYDRATION



H001 ｜ Observation → Literature → Mechanism

Question ｜提升 bulk conductivity 是否足以降低位置依賴的訊號變異？

Observation ｜位置依賴缺陷與訊號變異已觀察到。

H001 ｜ Literature → Mechanism → Strategy

Consensus ｜ transport gradient 可產生位置效應。
Alternatives ｜ interface contact remains alternative
Gap ｜缺少控制比較隔離機制。
Implication ｜需匹配條件後比較。

Mechanism ｜ bulk transport
Evidence ｜ C102; C103; E103

H001 ｜ Experiment Design

IV ｜含水量
Controls ｜電極幾何；接觸壓力
Baseline ｜原始配方
Sample ｜ replicates: 3; samples: 15
N ｜ 3
Outputs ｜導電度 (mS/cm)
Units ｜導電度 (mS/cm)
Method ｜ synthetic-four-probe

Prediction ｜導電度提升
Rule ｜ go: mean conductivity increases >=20%; partial_go: 10-20%; no_go: <10%

H001 ｜ Experiment Design

IV ｜位置
Controls ｜配方； 電極； 接觸壓力
Baseline ｜原始位置分布
Sample ｜ replicates: 3; samples: 15
N ｜ 3
Outputs ｜訊號 CV (%)
Units ｜訊號 CV (%)
Method ｜ synthetic-signal-test

Prediction ｜ CV 下降
Rule ｜ go: CV decreases >=30%;
partial_go: 10-30%; no_go: <10%

H001 │ Result

ID │ RES101
Result │ 平均導電度增加 24% ± 5% SD
Metric │ {'name': 'CV', 'value': 24, 'uncertainty': 5, 'units': '%'}

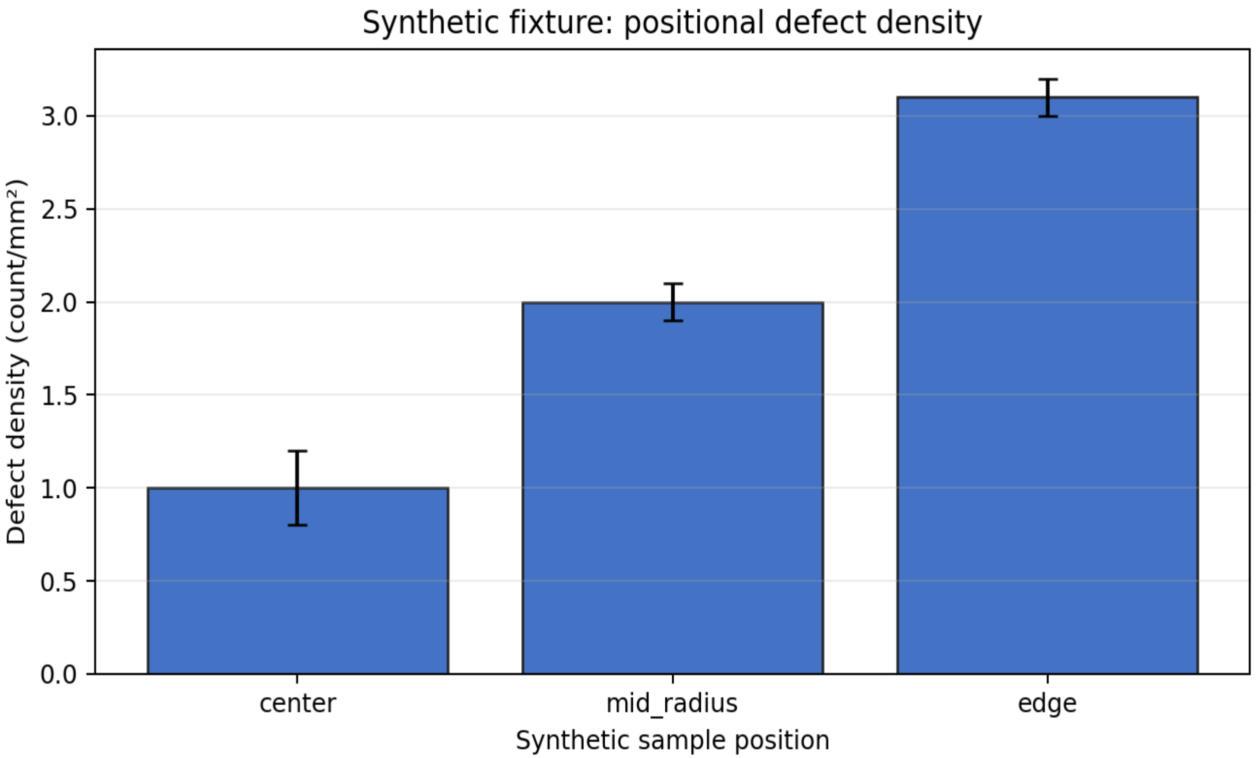


Figure │ RES101 │ 平均導電度增加 24% ± 5% SD

H001 ｜ Result

ID ｜ RES102
Result ｜訊號 CV 僅下降 4% ± 6% SD ，屬 No-Go
Metric ｜ {'name': 'CV', 'value': 4, 'uncertainty': 5, 'units': '%'}

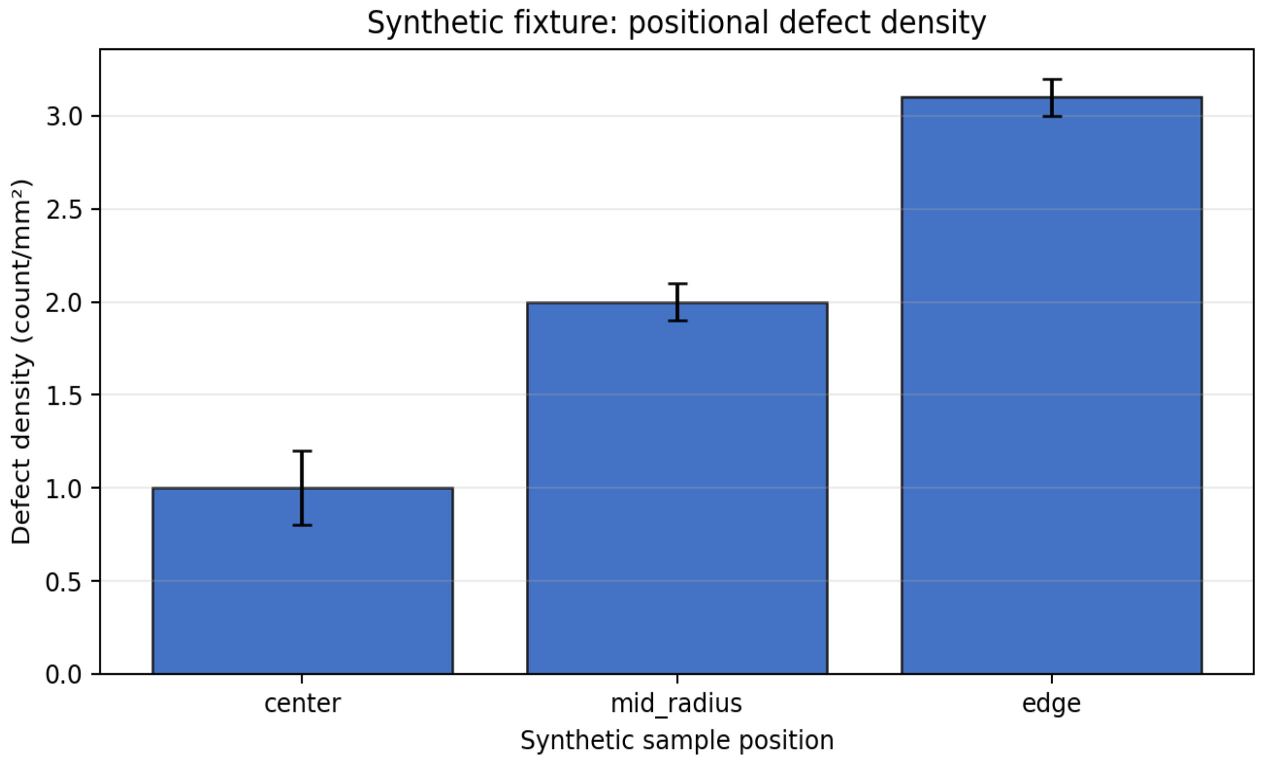


Figure ｜ RES102 ｜訊號 CV 僅下降 4% ± 6% SD ，屬 No-Go

H001 ｜ Integrated Discussion

Support ｜ RES101

Conflict ｜ RES102
Non-disc. ｜ none

Pattern ｜導電度提升成立，但重複性沒有同步改善。
Mechanism ｜ bulk conductivity 是部分機制，不足以解釋 interface-sensitive variation 。
Alternatives ｜接觸電阻；局部壓力差；電極幾何
Uncertainty ｜ contact interface 是否為主導因素仍需匹配導電度後檢驗。

H001 ｜ Layer Summary / Decision

Answer ｜ bulk conductivity 影響缺陷梯度，但不足以單獨解釋重複性。
Status ｜ partial_support
Decision ｜ Partial-Go; Bulk conductivity is contributory but insufficient.

Unresolved ｜ interface contact 的貢獻尚未量化。

Next question ｜匹配 bulk conductivity 後， contact resistance 是否主導變異？
Next step ｜ NS101

H001 ｜ Hypothesis Transition

Prior H ｜ C101
Key results ｜ RES101; RES102
Precursor ｜ E104
New H ｜ C201

Unresolved ｜導電度增加後訊號重複性沒有改善。
Rationale ｜剩餘變異隨電極接觸位置改變，因此核心解釋由 bulk transport 轉向 interface contact 。

H002 ｜ Hypothesis

Hypothesis ｜ Contact resistance drives instability at low pressure.
Falsifier ｜ Matched conditions do not change the predicted CV or resistance.
Question ｜低接觸壓力下， contact resistance 是否主導訊號不穩定？

H002 ｜ Problem

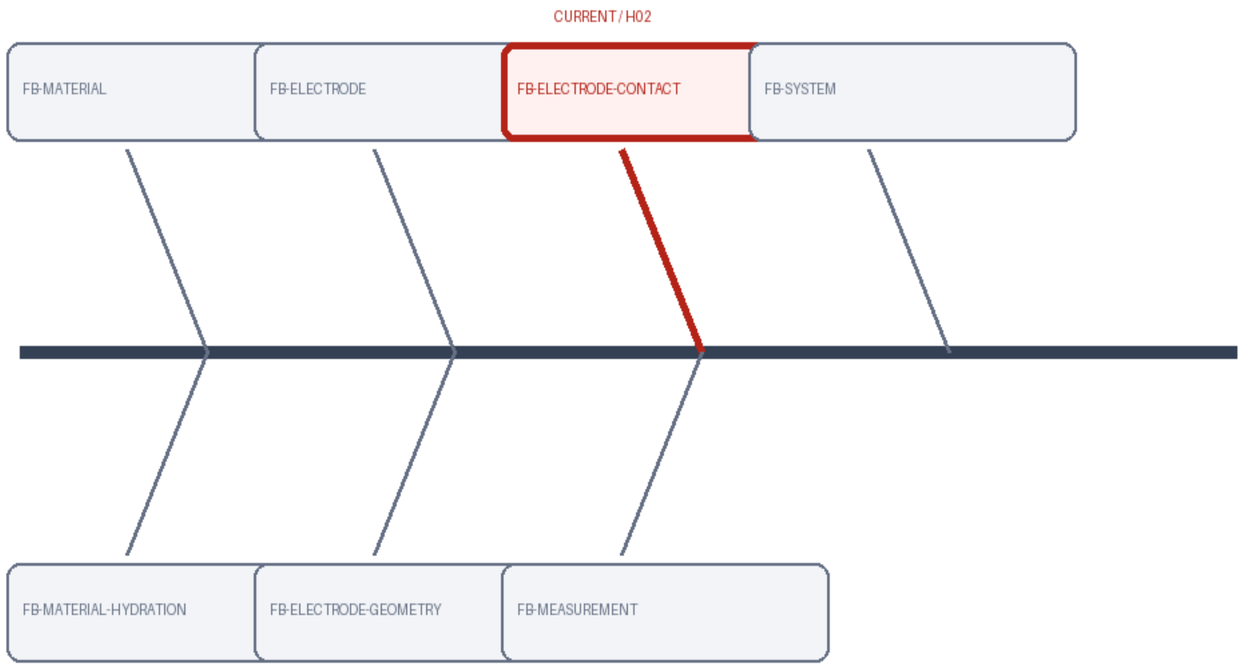
Prior ｜ H01 支持導電度梯度存在 ; H01 未支持導電度單獨主導重複性

Conflict ｜ bulk transport 與 interface contact 的相對貢獻尚未區分。

Question ｜低接觸壓力下，contact resistance 是否主導訊號不穩定？

H002 | Total Fishbone / Research Map

History | FB001 rev2
Focus | FB-ELECTRODE-CONTACT



H002 ｜ Observation → Literature → Mechanism

Observation ｜位置依賴缺陷與訊號變異已觀察到。

Consensus ｜ transport gradient 可產生位置效應。
Alternatives ｜ interface contact remains alternative
Gap ｜缺少控制比較隔離機制。
Implication ｜需匹配條件後比較。

Mechanism ｜ contact resistance
Evidence ｜ C202; C203; E202

Strategy ｜匹配導電度並改變接觸壓力
Criterion ｜預測指標跨過 decision threshold

Question ｜低接觸壓力下，contact resistance 是否主導訊號不穩定？

Registered observation visual.

H002 ｜ Results

IV ｜ contact pressure
Controls ｜ bulk conductivity; 電極幾何
Baseline ｜ low-pressure control
Sample ｜ replicates: 5; samples: 15
N ｜ 5
Outputs ｜訊號 CV (%); contact resistance (ohm)
Units ｜訊號 CV (%); contact resistance (ohm)
Method ｜ synthetic-pressure-fixture

Prediction ｜ CV and resistance decrease
Rule ｜ go: both metrics improve; partial_go: one metric improves; no_go: neither improves

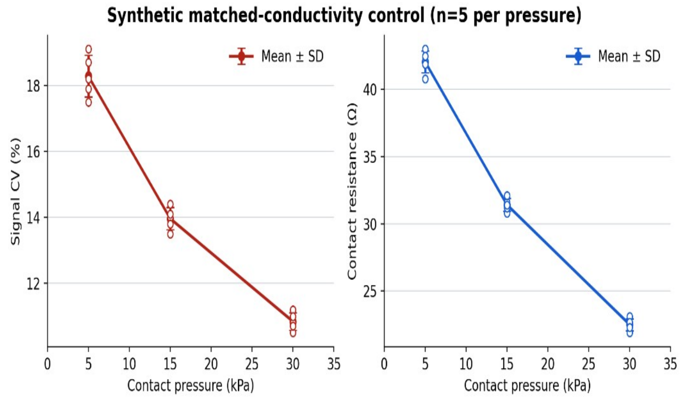


Figure ｜ RES201 ｜高壓條件 CV 下降 38% ± 7% SD ， contact resistance 同步下降

ID ｜ RES201
Result ｜高壓條件 CV 下降 38% ± 7% SD ， contact resistance 同步下降
Metric ｜ {'name': 'CV', 'value': 24, 'uncertainty': 5, 'units': '%'}

H002 ｜ Layer Summary / Decision

Support ｜ RES201

Conflict ｜ none
Non-disc. ｜ none

Cross-experiment pattern ｜匹配導電度後，提高接觸壓力明顯降低變異。
Mechanism assessment ｜ contact resistance 在低壓條件下主導不穩定。
Alternatives ｜電極幾何仍可能調節壓力分布

Pattern ｜匹配導電度後，提高接觸壓力明顯降低變異。
Mechanism ｜ contact resistance 在低壓條件下主導不穩定。
Alternatives ｜電極幾何仍可能調節壓力分布
Uncertainty ｜高壓長期循環是否造成新的 wear mechanism 尚未知。
Unresolved ｜高壓循環的耐久性與 wear 尚未測試。

Answer ｜低接觸壓力下 contact resistance 是主導不穩定來源。
Status ｜ supported
Decision ｜ Go; Contact-pressure control discriminates the interface mechanism.

Next question ｜接觸壓力改善能否在長期循環下維持？
Next step ｜ NS201